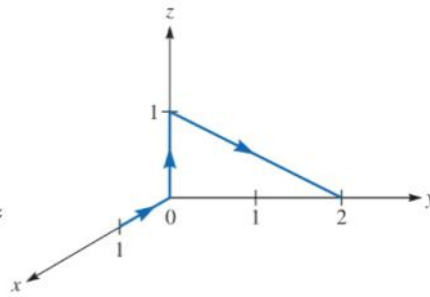


Activity 3

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Q1: If $\mathbf{H} = (x - y)\mathbf{a}_x + (x^2 + zy)\mathbf{a}_y + 5yz\mathbf{a}_z$
evaluate $\int_L \mathbf{H} \cdot d\mathbf{l}$ along the contour of Figure 1.



- $P_1 = (1, 0, 0)$
 $P_2 = (0, 0, 1)$
 $P_3 = (0, 2, 0)$

So,

$$\mathbf{H} = (x - y)\mathbf{a}_x + (x^2 + zy)\mathbf{a}_y + 5yz\mathbf{a}_z$$

$$\int_L \mathbf{H} \cdot d\mathbf{l} = I_1 + I_2 + I_3$$

- Path $P_1(1, 0, 0) \rightarrow P_2(0, 0, 1)$

$y = 0$, line in xz -plane connecting $(1, 0)$
to $(0, 1)$ so,

$$z = 1 - x, \quad dz = -dx$$

$$d\mathbf{l} = dx\mathbf{a}_x + dz\mathbf{a}_z = dx\mathbf{a}_x - dx\mathbf{a}_z$$

$$\therefore y = 0: \mathbf{H} = x\mathbf{a}_x + x^2\mathbf{a}_y + 0\mathbf{a}_z$$

$$\mathbf{H} \cdot d\mathbf{l} = x dx$$

$$I_1 = \int_1^0 x \cdot dx = \left. \frac{x^2}{2} \right|_1^0 = -\frac{1}{2}$$

• Path 2: $P_2(0,0,1) \rightarrow P_3(0,2,0)$

$x=0$, line in yz plane

$$z = 1 - y/2, \quad dz = -1/2 dy$$

$$d\mathbf{l} = dy \mathbf{a}_y + dz \mathbf{a}_z = dy \mathbf{a}_y - 1/2 dy \mathbf{a}_z$$

$$= \left(1 - y/2\right) y \mathbf{a}_y + 5y \left(1 - \frac{y}{2}\right) \mathbf{a}_z$$

$$H \cdot d\mathbf{l} = \left(y - \frac{y^2}{2}\right) dy + 5 \left(y - \frac{y^2}{2}\right) (-1/2) dy$$

$$= -3/2 \left(y - \frac{y^2}{2}\right) dy$$

$$I_2 = -3/2 \int_0^2 \left(y - \frac{y^2}{2}\right) dy = -3/2 \left[\frac{y^2}{2} - \frac{y^3}{6} \right]_0^2$$

$$= -3/2 \left(2 - \frac{8}{6}\right) = -3/2 \left(\frac{2}{3}\right) = -1$$

• Path 3: $P_3(0,2,0) \rightarrow P_1(1,0,0)$

$z=0$, line in xy -plane connecting $(0,2)$ to $(1,0)$. so,

$$y = 2 - 2x, \quad dy = -2dx$$

$$d\mathbf{l} = dx \mathbf{a}_x + dy \mathbf{a}_y = dx \mathbf{a}_x - 2dx \mathbf{a}_y.$$

$$\text{since } z=0: H = (x - (2 - 2x)) \mathbf{a}_x + x^2 \mathbf{a}_y + \mathbf{a}_z \cdot 0$$

$$= (3x - 2) \mathbf{a}_x + x^2 \mathbf{a}_y$$

- $H.dl = (3x-2) dx - 2x^2 dx = (-2x^2 + 3x - 2) dx$

- $I_3 = \int_0^1 (-2x^2 + 3x - 2) dx$

$$= \left[-\frac{2}{3}x^3 + \frac{3}{2}x^2 - 2x \right]_0^1 = -\frac{2}{3} + \frac{3}{2} - 2 = -\frac{7}{6}$$

- Total integral:

$$= I_1 + I_2 + I_3 = -\frac{1}{2} - 1 - \frac{7}{6} = \boxed{-\frac{8}{3}}$$

Q2: A vector field is specified as $\mathbf{A} = r\mathbf{a}_r - 3\mathbf{a}_\theta + 5\phi\mathbf{a}_\phi$.
Find the flux of the field out of the closed surface defined by
 $0 < r < 4, 0 < \theta < \pi/2, 0 < \phi < 3 < \pi/2$.

by divergence thm:

$$\oint \mathbf{A} \cdot d\mathbf{S} = \int_V (\nabla \cdot \mathbf{A}) dv$$

so,

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

- $A_r = r \rightarrow \frac{1}{r^2} \frac{\partial}{\partial r} (r^2) = \frac{3r^2}{r^2} = 3$

- $A_\theta = -3 \rightarrow \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (-3 \sin \theta) = \frac{-3 \cos \theta}{r \sin \theta} = -\frac{3 \cot \theta}{r}$

- $A_\phi = 5\phi \rightarrow \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (5\phi) = \frac{5}{r \sin \theta}$

$$\bullet \text{H}\phi = 5\phi \rightarrow \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (5\phi) = \frac{5}{r \sin \theta}$$

$$\nabla \cdot A = 3 - \frac{3 \cot \theta}{r} + \frac{5}{r \sin \theta}$$

Now, integrating it over $dv = r^2 \sin \theta \, dr \, d\theta \, d\phi$

$$\int_V (\nabla \cdot A) \cdot dv = \int_0^{\pi/2} d\phi \int_0^{\pi/2} \sin \theta \, d\theta \int_0^4 \left(3 - \frac{3 \cot \theta}{r} + \frac{5}{r \sin \theta} \right) r^2 \, dr$$

term #1:

$$\int_0^{\pi/2} d\phi \int_0^{\pi/2} \sin \theta \, d\theta \int_0^4 3r^2 \, dr = \left(\frac{\pi}{2} \right) (r^3)_0^4 = \frac{\pi}{2} \cdot 64 = 32\pi$$

term #2:

$$\int_0^{\pi/2} d\phi \int_0^{\pi/2} \cos \theta \cdot d\theta \int_0^4 (-3r) \, dr = \left(\frac{\pi}{2} \right) \left(\frac{-3r^2}{2} \right)_0^4 = -12\pi$$

term #3:

$$\int_0^{\pi/2} d\phi \int_0^{\pi/2} d\theta \int_0^4 5r \, dr = \frac{\pi}{4} \left(\frac{5r^2}{2} \Big|_0^4 \right) = \frac{\pi^2}{4} \cdot 40 = 10\pi^2$$

Now,

$$\text{Flux} = 32\pi - 12\pi + 10\pi^2 = \boxed{161.53}$$

Q3: Let $\mathbf{D} = 2\rho z^2 \mathbf{a}_\rho + \rho \cos^2 \phi \mathbf{a}_z$. Evaluate

(a) $\oint_S \mathbf{D} \cdot d\mathbf{S}$

(b) $\int_V \nabla \cdot \mathbf{D} dv$

over the region defined by $2 \leq \rho \leq 5, -1 \leq z \leq 1, 0 < \phi < 2\pi$.

in cylindrical coordinates:

$$\nabla \cdot \mathbf{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

b)

$$\bullet D_\rho = 2\rho z^2 \rightarrow \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) = 4z^2$$

$$\bullet D_\phi = 0$$

$$\bullet D_z = \rho \cos^2 \phi \rightarrow \frac{\partial}{\partial z} (\rho \cos^2 \phi) = 0$$

so,

$$\nabla \cdot \mathbf{D} = 4z^2$$

Now,

$$\int_V \nabla \cdot \mathbf{D} dv = \int_{-1}^1 4z^2 dz \int_0^{2\pi} d\phi \int_2^5 \rho d\rho$$

$$\bullet \int_{-1}^1 4z^2 dz = \left. \frac{4z^3}{3} \right|_{-1}^1 = 8/3$$

$$\bullet \int_0^{2\pi} d\phi = 2\pi$$

$$\bullet \int_2^5 \rho \, d\rho = \frac{\rho^2}{2} \Big|_2^5 = 21/2$$

$$\int_V \nabla \cdot \mathbf{D} \, dv = \frac{8}{3} \times 2\pi \times \frac{21}{2} = 56\pi$$

a)

1- Outer cylinder ($\rho=5$): $ds = 5 \, d\phi \, dz \, \mathbf{a}_\rho$

$$\begin{aligned} \iint \mathbf{D} \cdot d\mathbf{s} &= \int_{-1}^1 \int_0^{2\pi} 10 \, z^2 (5) \, d\phi \, dz = 50 \int_{-1}^1 z^2 \, dz \int_0^{2\pi} d\phi \\ &= 50 \left(\frac{2}{3} \right) (2\pi) = \frac{200\pi}{3} \end{aligned}$$

2- Inner cylinder ($\rho=2$) $ds = -2 \, d\phi \, dz \, \mathbf{a}_\rho$

$$\iint \mathbf{D} \cdot d\mathbf{s} = \int_{-1}^1 \int_0^{2\pi} 4 \, z^2 (-2) \, d\phi \, dz = -8 \left(\frac{2}{3} \right) (2\pi) = \frac{-32\pi}{3}$$

3- Top disk ($z=1$) $ds = \rho \, d\rho \, d\phi \, \mathbf{a}_z$

$$\begin{aligned} \mathbf{D} \cdot d\mathbf{s} &= \int_2^5 \int_0^{2\pi} (\rho \cos^2 \phi) \rho \, d\rho \, d\phi = \int_2^5 \rho^2 \, d\rho \int_0^{2\pi} \cos^2 \phi \, d\phi \\ &= \frac{\rho^3}{3} \Big|_2^5 (\pi) = \frac{125-8}{3} \pi = 39\pi \end{aligned}$$

4- Bottom disk ($z=-1$): $ds = -\rho \, d\rho \, d\phi \, \mathbf{a}_z$

$$\iint D \cdot ds = \int_2^5 \int_0^{2\pi} (p \cos^2 \phi)(-p) dp d\phi = -39\pi$$

$$\text{Total} = \frac{200\pi}{3} - \frac{32\pi}{3} + 39\pi - 39\pi = 56\pi$$

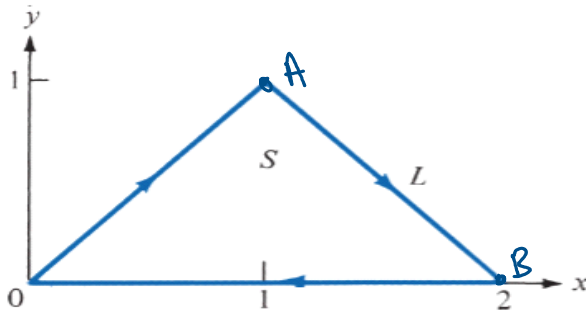
Q4:

Given that $\mathbf{F} = x^2 y \mathbf{a}_x - y \mathbf{a}_y$, find

(a) $\oint_L \mathbf{F} \cdot d\mathbf{l}$, where L is shown in Figure 2.

(b) $\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where S is the area bounded by L .

(c) Is Stokes's theorem satisfied?



1) $O(0,0)$, $A(1,1)$, $B(2,0)$

• segment 1: $O(0,0) \rightarrow A(1,1)$

$$y=x \rightarrow dy=dx \quad x \in [0,1]$$

$$\bullet \quad \mathbf{F} \cdot d\mathbf{l} = x^2 y dx - y dy = x^3 dx - x dx = (x^3 - x) \cdot dx$$

$$\bullet \quad I_1 = \int_0^1 (x^3 - x) dx = \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_0^1 = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$$

• Segment 2: $A(1,1) \rightarrow B(2,0)$

line equation $y=2-x \rightarrow dy=-dx$ mit $x \in [1,2]$

$$F \cdot dl = x^2 (2-x) dx - (2-x) (-dx) = (2x^2 - x^3 + 2 - x) dx$$

$$\bullet I_2 = \int_1^2 (-x^3 + 2x^2 - x + 2) dx = \left[\frac{-x^4}{4} + \frac{2x^3}{3} - \frac{x^2}{2} + 2x \right]_1^2$$

at $x=2$:

$$-4 + 16/3 - 2 + 4 = 10/3$$

at $x=1$:

$$-1/4 + 2/3 - 1/2 + 2 = 23/12$$

$$I_2 = 10/3 - 23/12 = 17/12$$

• Segment 3: $B(2,0) \rightarrow O(0,0)$

$$y=0 \rightarrow dy=0$$

$$F=0 \rightarrow I_3=0$$

• Total line integral:

$$\oint_L F \cdot dl = -1/4 + 17/12 + 0 = 14/12 = \boxed{7/6}$$

b)

calculating curl:

$$\nabla \times F = \begin{vmatrix} \partial_x & \partial_y & \partial_z \end{vmatrix} = \begin{vmatrix} 0 & -x(x^2y) & 0 \end{vmatrix} = -x^2 \partial_z$$

$$\nabla \times F = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & -y & 0 \end{vmatrix} = \left(0 - \frac{\partial(x^2y)}{\partial x} \right) \mathbf{a}_z = -x^2 \mathbf{a}_z$$

using surface normal $ds = dx dy \mathbf{a}_z$:

$$\int_S (\nabla \times F) ds = \iint_S -x^2 dx dy$$

Now split triangle:

$$1. x \in [0, 1], y \in [0, x]$$

$$\int_0^1 \int_0^x -x^2 dy dx = \int_0^1 -x^3 dx = \left[-\frac{x^4}{4} \right]_0^1 = -\frac{1}{4}$$

$$2. x \in [1, 2], y \in [0, 2-x]:$$

$$\begin{aligned} \int_1^2 \int_0^{2-x} -x^2 dy dx &= \int_1^2 -x^2 (2-x) dx = \int_1^2 (x^2 - 2x^2) dx \\ &= \left[\frac{x^3}{3} - \frac{2x^3}{3} \right]_1^2 \end{aligned}$$

$$\text{at } x=2: 4 - \frac{16}{3} = -\frac{4}{3}$$

$$\text{at } x=1: \frac{1}{3} - \frac{2}{3} = -\frac{1}{3}$$

$$\text{value: } -\frac{4}{3} - \left(-\frac{1}{3} \right) = -\frac{3}{3} = -1$$

$$\text{adding both: } -\frac{1}{4} - \frac{11}{12} = \boxed{-\frac{7}{6}}$$

-ve due to direction.

adding both. $-\frac{1}{4} - \frac{11}{12} = \boxed{-\frac{7}{6}}$ ✓

c) Yes the Stokes theorem is satisfied as the closed line integral equals the surface integral of the curl of the vector field over the bounded region.

$$\oint_C \mathbf{F} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$